INCH-POUND

MIL-DTL-32693/1
30 November 2021

DETAIL SPECIFICATION SHEET

CONVERTER, DIRECT CURRENT TO ALTERNATING CURRENT, SINGLE PHASE, 60 HERTZ, MULTI-PURPOSE, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32693.

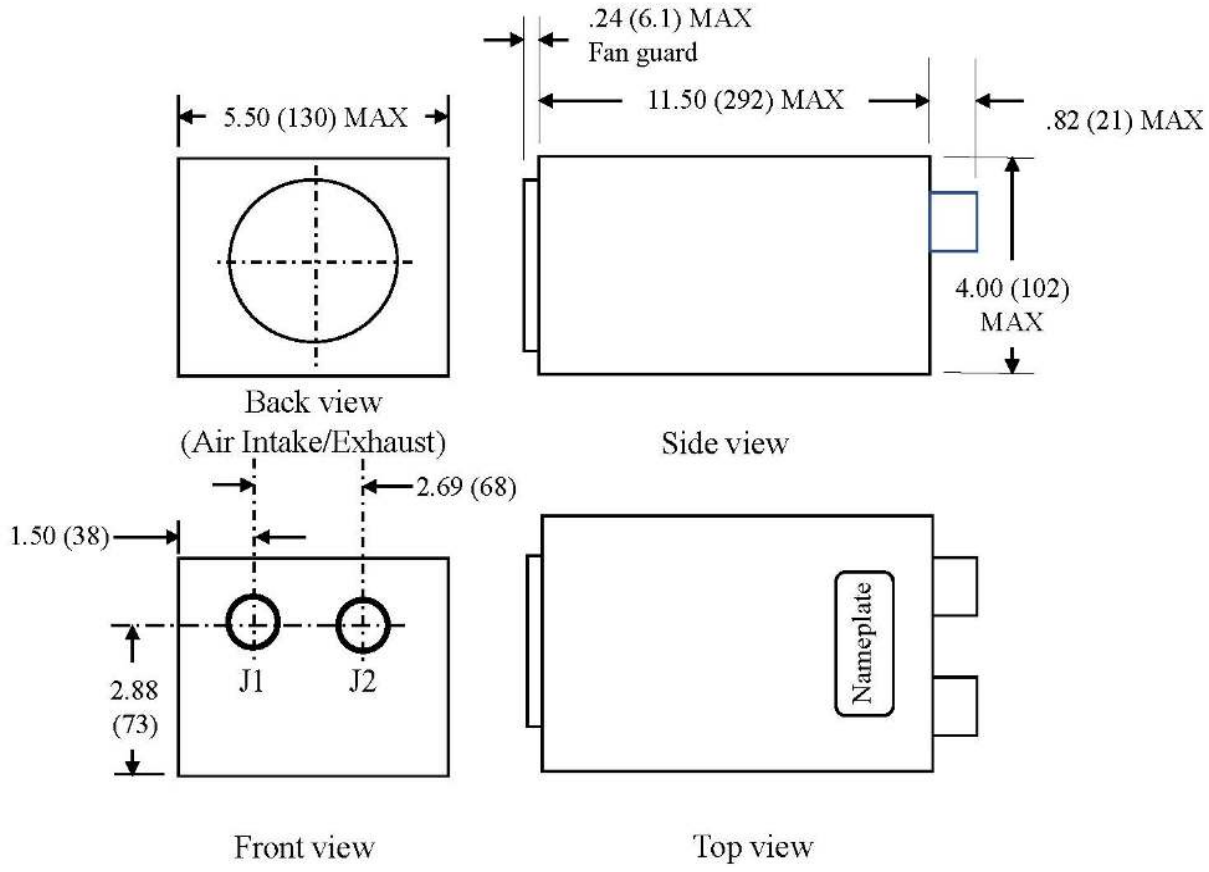


FIGURE 1. Static converter M32693/1-1.

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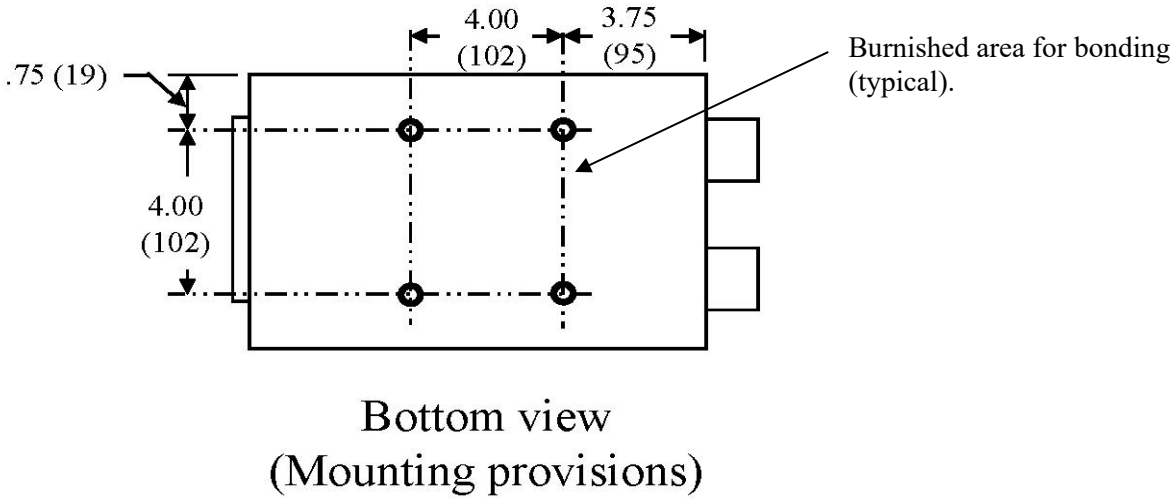


FIGURE 1. Static converter M32693/1-1 – Continued.

NOTES:

1. Unless otherwise specified, dimensions are in inches: fractional tolerance ± 1/16; decimal tolerances ± 0.1 for one decimal (0.0), ± 0.03 for two decimals, ± 0.010 for three decimals; angular tolerances ± 1/2°.

2. Metric equivalents are shown in parenthesis, are for reference only. Conversions given in this document are based on the following formulas:

   a. Millimeters = the number of inches * 25.4 millimeters/inch.

   b. °F = 1.8 * temperature in °C + 32.

   c. Length in kilometers = distance in feet * .3048 meters/foot * 1 kilometer /1000 meters.

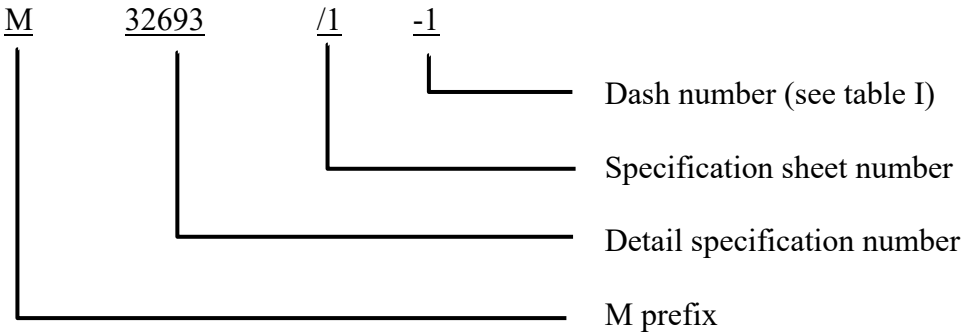
REQUIREMENTS

1. Converter type. The converter type shall be Type DC.

2. Temperature/altitude rating. The converter shall provide the electrical performance over a temperature altitude range of -55 °C to + 71 °C (-67 °F to 160 °F) in accordance with MIL-DTL-32693, figure 9 up to 20,000 feet (6.1 km) altitude and as modified herein. The maximum operating altitude is 20,000 feet (6.1 km).

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3. Part or identifying number (PIN) example:



4. Converter marking. The converter nameplate shall be located on the top of the converter, as shown on figure 1. When an integral cooling fan is used, the direction of airflow shall be plainly and permanently marked on the outside of the housing in the direction specified on figure 1.

5. Converter rating:

TABLE I. Converter description.

Part number	Power rating 1/	Power factor	Max weight lb. (kg)	Efficiency
M32693/1-1	500VA	0.75 Lagging (Min) 0.85 Leading (Min)	9 (4)	75% minimum

1/ Single phase.

6. External electric connections.

TABLE II. Connector contact assignments.

Connector	Contact	Description
J1 (Input) Pin contacts	A	Spare
	B	28 VDC (Volts Direct Current) Return
	C	Spare
	D	+28 VDC
	E	Spare
	F	Spare
J2 (Output) Socket contacts	A	Phase A Output
	B	Spare
	C	Spare
	D	Output Return
	E	Chassis Ground
	F	Chassis Ground
	G	Spare
	H	Spare
	J	Signal Ground

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TABLE II. Connector contact assignments – Continued.

Connector	Contact	Description
J2 (Output) Socket contacts	K	Spare
	L	Phase A Output
	M	Spare
	N	Spare
	P	Spare
	R	Output Return
	S	Spare
	T	Spare
	U	Spare

7. Connectors. Connector J1 shall mate with D38999/26WE6SN as specified in MIL-DTL-38999/26 and oriented as shown on figure 1. Connector J2 shall mate with D38999/26WD18PN and oriented as shown on figure 1. The input and output connections and contact assignments shall be marked, as required in table II and on figure 1.

8. Converter mounting. The converter shall be mounted by four 10-32 UNF-3A screws. Provisions shall be made in the bottom of the converter for four by four threaded mounting holes as shown in figure 1 to mate with the four 10-32 UNF-3A screws. The depth of the mounting holes shall be a minimum of .43 inch (11 mm). The converter, when mounted by the mounting screws, shall meet the vibration and shock requirements of this detail specification sheet.

9. Cooling. The use of an internally mounted, an integral cooling fan is permitted.

10. Ventilation openings. All ventilation openings shall be designed to prevent the passage of foreign objects and shall be of such size to prevent the passage of objects of 1/4 inch (6.35 mm) or larger.

11. Size and weight. The size and dimensions of the converter shall be as shown on figure 1. Converter weight shall be as required in table I.

12. Chassis ground (Contacts E and F of J2). The Direct Current (DC) resistance from the connector contacts E and F to the converter's chassis shall not exceed 50 milliohms.

13. Current returns. The Alternating Current (AC) neutral and DC current return terminals shall be connected to its own current return wire. The current return wiring shall be separate from any wires used to bond internal components to the chassis of the converter.

14. Type AC converter input power. Not required.

15. Overload and short circuit capacity. Over an input voltage range of 22 to 29 VDC, the converter shall be capable of operating without damage into any passive load impedance (including a short circuit) for five seconds.

a. Overload. Over the Type DC normal input voltage range, the converter shall, without damage, maintain the minimum output voltage while delivering 150 percent rated output current for 15 seconds.

16. Efficiency. See table I.

17. Electromagnetic compatibility. The converter shall conform to the requirements of

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MIL-STD-461 and MIL-STD-464, when tested to the requirements of this detail specification sheet.

18. Electrical bonding. Electrical bonding of the converter shall be through burnished areas around the four mounting holes of the converter (see figure 1). The bonding resistance between the mounting surface and any part of the converter shall comply with the electrical bonding requirements of MIL-STD-464. The bonding resistance between the mounting surface and each connector case shall be not greater than 2.5 milliohms.

19. Safety. Converter safety design and operational safety requirements shall be in accordance with those applicable requisites presented in the MIL-STD-882, system safety process section.

20. Acoustical noise. Not required.

VERIFICATION:

1. Airflow. Not applicable.

2. Converter input power. Only type DC input power required. Type AC input power is not required.

3. Input power factor. Not applicable.

4. Input current balance. Not applicable.

5. Input starting current. Not applicable.

6. Total harmonic distortion. Not applicable.

7. Output load recovery. For 200 percent load and short circuit application and removal, tests shall be conducted at nominal input voltage, 22 VDC input, and 29 VDC input.

8. Electromagnetic compatibility. The converter shall be tested and be in accordance with the requirements and applicable limits of MIL-STD-461. All converters shall meet the requirements of CE102, CS101, CS114, CS115, CS116, CS117, CS118, RE102, and RS103. All output leads, including the neutral, shall comply with the requirements of CE102. Converters used in Army or Anti-submarine Warfare (ASW) applications shall also meet the requirements of CE101, RE101, and RS101. The test shall be conducted at full load and half load, at minimum and maximum normal input voltage. The converter load power factor shall be adjusted such that the combined power factor of the load and all auxiliary test equipment required shall present a total load of unity power factor to the converter. During electromagnetic susceptibility testing, the output voltage and frequency shall remain within the normal limits (see MIL-DTL-32693, 3.5.1.3.1.1). Additionally, the converter shall comply with the electrical bonding requirements of MIL-STD-464.

9. External electromagnetic environment. As installed in the aircraft, the output voltage and frequency shall remain within the normal limits (see MIL-DTL-32693, 3.5.1.3.1.1) while the converter is exposed to a worst-case composite of the MIL-STD-464 maximum external electromagnetic environments for deck operations, ship operations in the main beam of transmitters, and rotary-wing aircraft, excluding shipboard operations.

10. Input voltage unbalance. Not applicable.

11. Input waveform distortion. Not applicable.

12. Transient input frequency. Not applicable.

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13. Acoustical noise. Not applicable.

14. Acceleration. The converter shall be tested in accordance with the acceleration test of MIL-STD-810, Method 513, Procedure II. The required acceleration level for all directions in procedure II (operational test) shall be 6.0g and shall be applied along three mutually perpendicular axes in two opposite directions along each axis. The test time duration in each direction shall be at least one minute following centrifuge stabilization, and the converter shall be operated continuously at 28 VDC input and full load during the test. The output voltage and frequency shall be monitored during the test to determine compliance with MIL-DTL-32693, 3.5.2.2. At the conclusion of the test, the converter shall be examined for evidence of damage, and voltage and frequency regulation shall be measured as specified in MIL-DTL-32693, 4.6.2.1.

15. Shock. Applicable. In addition, calibration shocks shall be applied in accordance with MIL-STD-810, Method 516, prior to shock testing.

a. Functional. The converter shall be operating during the functional shock test at full load and 28 VDC input. The output voltage shall be monitored during the test to determine compliance with MIL-DTL-32693, 3.5.2.4a. At the conclusion of the functional shock test, the converter shall be inspected for evidence of damage. Voltage and frequency regulation shall be measured as specified in MIL-DTL-32693, 4.6.2.1.

b. Crash hazard. The converter shall not be operated during the crash hazard shock test. At the conclusion of the crash hazard shock test, there shall be no failure of the mounting attachment, the converter being tested shall remain in place, and no parts shall have separated from the converter. However, bending and distortion of the converter shall be permitted.

16. Vibration. Delete in its entirety and replace with the following: “The converter shall be subjected to vibration testing of MIL-STD-810, Method 514, Procedure I (general vibration), Annex D, Category 14 (rotary-wing aircraft - helicopter). The converter shall be operating during the vibration test at full load and 28 VDC input. The output voltage and frequency shall be monitored during the test to determine compliance with MIL-DTL-32693, 3.5.2.5. The applied vibration reference profile shall be MIL-STD-810, Method 514.8, Annex D, Category 14, helicopter vibration exposure requirements (general materiel) tailored to the AH-1 aircraft platform using the sinusoidal frequencies and magnitudes shown in table III of this detail specification sheet. The duration shall be 4 hours in each of three mutually perpendicular axes. Equivalence methods may be applied to test durations where applicable (i.e., amplitudes may be reduced and durations increased) per guidance provided in MIL-STD-810, Method 514.8, Annex F. The test article shall demonstrate specific performance during and upon completion of this test sequence and exhibit no structural damage or degradation. Pre- and post-test frequency response measurements shall be used to identify structural damage.”

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TABLE III. Sine-on-random profile parameters.

Description	Hz	Amplitude
Main Rotor 1P (A_1 /f_1)	4.9	0.12 g
Main Rotor Blade Pass (A_2 /f_2)	19.7	1.97 g
2X Main Rotor Blade Pass (A_3 /f_3)	39.4	2.50 g
3X Main Rotor Blade Pass (A_4 /f_4)	118.1	1.50 g
W_0		0.010 g^2/Hz
W_1		0.010 g^2/Hz
f_t	500	

17. Resonance search. Not applicable.

18. Sinusoidal vibration test. Not applicable.

19. Referenced documents. In addition to MIL-DTL-32693, this detail specification sheet references MIL-STD-461, MIL-STD-464, MIL-STD-810, MIL-STD-882 and MIL-DTL-38999/26.

CONCLUDING MATERIAL

Custodians:
Navy - AS
DLA - GS

Preparing activity:
Navy - AS

Project 6130-2021-002

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